TAP CHANGING
RANSFORMERS

TAP CHANGERS

es for changing the voltage of a transformer.

ry to maintain bus voltages on the secondary side

vels.

ange is obtained by changing number of turns in

l secondary windings.

p changers are available:

ap Changer :

per.

g is effected when the transformer is off

ap changers are used where tap changing is done ay as in the case of a Distribution transformer and es for Generator Transformers.

fter each tap change, the transformer has to be suitable megger for proper continuity at the new

ff load tap changer when transformer is in service extensive damage.

n electro magnetic latch switch is provided to dvertent operation.

changer :

hanging is done while transformer is in service nterrupting loads, as in the case of a power r.

changer normally consists of transition resistors ge the circuit between tapping section during the operation.

C TRANSFORMER

former is defined as a transformer that does not use nsulating or cooling medium for its windings or core. dings and core are enclosed in a sealed tank that is gas under pressure.

ormers are widely used in various industries and ause of their safety, reliability, and environmental n operate in harsh conditions such as high humidity, mic events without compromising their performance ard to people or property.

Dry type Transformer

s of Dry type Transformers

people and property because they do not contain any liquid that can leak or catch fire.

ce-free and pollution-free because they do not require tests, oil spills cleanup, or special disposal methods.

stall because they do not require any vaults or special an be placed close to the load to reduce the need for long ge cables.

entally friendly because they do not emit any harmful o the greenhouse effect.

llent capacity to support overloads because they have on and thermal endurance than oil-filled transformers.

es of Dry type Transformers

xpensive than oil-filled transformers for the same e rating because they require more materials and uring.

nd heavier than oil-filled transformers for the same ge rating because they have more air gaps and ss.

nsitive to dust, dirt, and vermin because they have hat can allow the entry of foreign particles that can ngs or cause short circuits.

than oil-filled transformers because they have iction and vibration that can produce audible sounds.

s of Dry type Transformers

s industry: Dry-type transformers are used to supply power to d processes that involve flammable or explosive substances, rochemical plants, pipelines, offshore platforms, etc.

itive areas: Dry-type transformers are used to protect the spills or leaks that can contaminate water sources, soil, or as water protection areas, forests, wetlands, etc.

ype transformers are used to prevent fire hazards or minimize hat are prone to fire outbreaks or have strict fire regulations, ations, underground substations, hospitals, schools, hotels,

n: Dry-type transformers are used to connect renewable e grid or to the load, such as wind turbines, solar panels, tc.

ry-type transformers are also used in other applications that ance, low maintenance, or special features, such as traction ms, mining systems, data centers, etc.